Ranked Execution: Cash-Funded Search for Sparse Portfolio Rebalancing

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Abstract

Implementing a target portfolio requires choosing trades and sizing positions under funding, exposure and minimum-trade constraints. We present Ranked Execution, a bounded heuristic combining cash-funded marginal priorities, joint convex sizing and coordinated changes of executable instrument states. Mandatory interventions are declared before risk search and count towards the trade limit. An incumbent is returned only after an audit in original portfolio units. A matched mixed-integer formulation provides independently started reference solutions under the same objective, prepared domain and acceptance rules. A portable benchmark is constructed from supplied model weights and covariance estimates for a retrospective universe of 302 publicly traded exchange-traded funds. It contains 219 monthly conditional decisions with opaque asset identifiers and requires neither an upstream allocation model nor private portfolio data. Search ablations and separate incumbent controls distinguish algorithmic improvement from external incumbent retention. The numerical study evaluates execution quality and computation conditional on the supplied inputs; it does not estimate investment performance or establish global optimality of the heuristic. All 219 cases deliver audited portfolios. Ranked matches Ranked-started Exact in all 219 cases within 10^{-4} annual basis points. Exact is faster on this cohort, whose zero to five optional candidates leave the ten-ticket allowance nonbinding.

Keywords: portfolio rebalancing, trade cardinality, minimum trade, tracking error, mixed-integer quadratic optimization

1. Introduction

A model portfolio describes desired exposures. Implementing every change can require many small orders, while selecting only the largest weight

changes can overlook their joint risk consequences. The implementation problem is therefore a discrete choice of trades followed by a continuous choice of sizes. Funding, group exposures, minimum trade sizes and minimum held positions couple these choices.

We study a conditional execution problem. Current holdings, target weights, a covariance estimate and execution constraints are supplied inputs. The task is to reduce tracking variance to the target using a limited number of actual noncash trades. A trade limit counts movements from the current book; it is distinct from a limit on terminal holdings. Cash is the settlement coordinate and need not be riskless. Required interventions, such as an exit or an upper-bound correction, consume tickets even when explicitly exempted from a minimum order size.

Ranked Execution evaluates cash-funded marginal directions and then jointly sizes selected executable states. Single-name improvements are supplemented by a bounded shortlist of coordinated state changes. These can admit a funding pair whose isolated trades are unhelpful, or expose another instrument after an earlier useful pair. Every accepted improvement passes the same original-unit audit. A native solver status alone does not establish portfolio acceptance.

The contribution has three parts. First, the formulation keeps original holdings, a prepared feasible base and raw model targets distinct. This separation makes mandatory accounting and minimum-ticket semantics explicit. Second, the search combines inexpensive covariance-aware priorities with joint sizing under the complete coupled constraints. Third, a portable benchmark and independent checks expose failures, search limits, timing and dependence on supplied incumbents.

The empirical input universe comprises 302 publicly traded ETFs. Model weights and covariances are frozen before execution comparisons. Numerical instances have opaque asset identifiers and contain no private fund observations. Readers do not need proprietary signals or a risk-model refit to reproduce execution. Each decision begins from the drifted previous model allocation, so every method receives identical current holdings. This experiment measures conditional implementation decisions, not strategies recursively evolving their own holdings.

A matched mixed-integer reference fixes the objective, admissible states and return audit. A separate call receives the heuristic's incumbent and charges its generation time explicitly. Two search ablations quantify incremental algorithmic effects. A historical production policy with different con-

straints is outside this comparison.

2. Related work

Cardinality-constrained portfolio selection and index tracking have a substantial optimization literature. Mutunge and Haugland (2018) analyse covariance tracking with a restricted number of holdings and study greedy construction and exchange improvement. This is a close methodological reference. Here cardinality applies to movements from an existing portfolio, with a distinguished funding coordinate and executable trade/holding intervals.

Restricted-support search is also established. Guastaroba and Speranza (2012) develop Kernel Search for mixed-integer index-tracking formulations by exploring restricted portions of the feasible region. Ranked Execution shares the principle of prioritising small subproblems. Its priority comes from one- and two-coordinate cash-funded quadratics; the subsequent subproblem sizes the entire selected support under coupled constraints.

Strub and Baumann (2018) formulate an index-tracking rebalancing problem that explicitly balances transaction costs and tracking accuracy. Our objective is ex-ante tracking variance with hard operating rules. Transaction-cost minimisation, market impact and execution scheduling are outside the model. A ticket cap is an operating constraint, not a calibrated replacement for every source of trading cost.

Bertsimas and Cory-Wright (2022) develop scalable outer approximation for sparse portfolio selection using ridge regularisation and a convex binary reformulation. This demonstrates the value of strong formulations and carefully designed computation. We do not claim that bounded local search supersedes those methods. Greedy selection, convex sizing and support exchange are established ideas; the present contribution combines them with conditional trade states, funding and audited delivery.

The empirical question is how well this specific combination approximates a matched mixed-integer solution under declared computational controls. Direct comparisons with additional established methods on the same trade domain remain an extension. Differences in objectives and constraints must be resolved before algorithms from other portfolio formulations can serve as matched controls.

3. Conditional execution problem

3.1. Inputs and objective

Let $c, m, w \in \mathbb{R}^N$ denote current, model and delivered weights, measured as fractions of net asset value. Index k identifies settlement cash and $\Sigma \succeq 0$ is the supplied annual covariance matrix. The objective and reported tracking error are

$$f(w) = (w - m)^\top \Sigma (w - m), \qquad \mathrm{TE}(w) = \sqrt{f(w)}. \tag{1}$$

Annual basis points equal $10^4\,\mathrm{TE}(w)$. No alpha term, turnover penalty or class-specific risk price enters this execution objective.

The long-only hard domain includes

$$\mathbf{1}^\top w = 1, \quad L \le w \le U, \quad a \le Aw \le b. \tag{2}$$

Group rows may be fractional: a mixed-asset instrument need not belong wholly to one group. Bounds and eligibility can encode calendar restrictions and instructions. Cash bounds are real constraints; funding through cash does not imply unlimited borrowing or unrestricted cash deployment.

A terminal noncash holding is zero or at least its declared floor:

$$w_i = 0 \quad \text{or} \quad w_i \ge \theta_i, \qquad i \ne k. \tag{3}$$

An ordinary discretionary movement satisfies

$$w_i = c_i \quad \text{or} \quad |w_i - c_i| \ge \delta_i. \tag{4}$$

The minimum applies to the delivered order, not merely to the desired model-current difference. Joint sizing can otherwise reduce an eligible desired trade to a much smaller actual movement.

Let $\tau_i(w) = \mathbf{1}\{w_i \ne c_i\}$ be the mathematical trade indicator. The total noncash count satisfies

$$\sum_{i \ne k} \tau_i(w) \le K. \tag{5}$$

Numerical delivery uses a separately declared trade-zero threshold. The model does not impose a terminal position-count limit.

3.2. Mandatory declarations and corridors

The consumer resolves lifecycle transitions, instructions and ordinary product bounds before risk search. Let q^0 be the resulting funded base. Every noncash base-current movement receives a mandatory category. A required correction can have a fixed endpoint or a one-sided interval allowing further movement in the permitted direction. A minimum-ticket exemption removes only the minimum-size condition: it does not remove the trade from the count or waive a holding floor.

For optional instrument i, an ordinary corridor lies between its base and target, intersected with eligible hard bounds:

$$[\ell_i, h_i] = [\min(q_i^0, m_i), \max(q_i^0, m_i)] \cap [L_i, U_i]. \tag{6}$$

The actual supplied intervals also retain lifecycle decisions, pins and declared preparation exceptions. Mandatory instruments keep their own resolved intervals. Cash retains its hard funding interval instead of a model-current corridor.

The eligible hard domain permits movement within full resolved product and instruction bounds. We call the two domains CORRIDOR and FULL. They answer different optimization questions. Every algorithm in the main comparison uses CORRIDOR. FULL is used by common preparation only when the ordinary corridor cannot supply a valid base.

If the policy declares a discretionary allowance B, the total cap is $K = M + B$, with M unique active mandatory names after preparation. This is an explicit convention. An input declaring a fixed total cap must retain it; preparation cannot silently enlarge that cap when declaring another repair.

3.3. Common feasibility preparation

Individual lifecycle decisions can produce a base that violates funding, group bands or holding floors. Separately repairing each algorithm's base would conflate preparation and risk-search effects. We prepare one shared instance before running any comparison arm.

An already audited base is retained exactly. Otherwise, preparation searches CORRIDOR and then, if needed, eligible FULL bounds. Its objectives are risk-neutral. In FULL, corridor excursion is minimised before gross noncash intervention and the number of additional repair instruments. Within the selected discrete pattern, a strictly convex squared-intervention

objective breaks continuous ties. Target risk and alpha do not choose among repairs.

Additional noncash interventions are pinned, declared and counted before risk search. Endpoints outside an original corridor are recorded as preparation exceptions, while full hard bounds and eligibility remain unchanged. All arms receive the same final base q, intervals, mandatory flags, exemptions and resolved cap.

Sequential optimization stages use an explicit numerical objective guard; they are not exact symbolic lexicographic arithmetic. A shared 30-second allowance includes preparation stages. Failed preparation remains in the experiment denominator. A numerical failure or exhausted budget alone does not establish mathematical infeasibility. Conditional solver comparisons state how many inputs successfully prepared.

4. Executable states and matched formulation

4.1. Finite unions of intervals

For each noncash instrument, compile executable alternatives $\mathcal{J}_i = \{[\ell_{ij}, h_{ij}] : j \in J_i\}$. For an ordinary optional instrument these include the no-trade point when valid, a sell interval ending at $c_i - \delta_i$, and a buy interval starting at $c_i + \delta_i$. Intersect the alternatives with the permitted corridor and with $\{0\} \cup [\theta_i, \infty)$, and remove empty intervals. Mandatory exemptions and fixed pins are applied before compiling the states.

Each state carries a ticket flag $t_{ij} \in \{0, 1\}$ measured from original holdings. An optional no-trade point has flag zero; an exempt mandatory movement keeps flag one. Once states are fixed, remaining sizing is a continuous convex problem.

For example, let $c_i = 0.02$, the corridor be $[0, 0.04]$, $\delta_i = 0.005$ and $\theta_i = 0.01$. The alternatives are a no-trade point at 0.02, a full exit at zero, a sell interval $[0.01, 0.015]$, and a buy interval $[0.025, 0.04]$. Selecting a sale does not prescribe its unique size. Funding and group constraints determine final size jointly with other names. Several states never create several tickets for one instrument.

4.2. Mixed-integer reference

Introduce binary selectors z_{ij} and continuous contributions v_{ij}:

$$\sum_{j \in J_i} z_{ij} = 1, \qquad \ell_{ij} z_{ij} \le v_{ij} \le h_{ij} z_{ij}, \tag{7}$$

$$w_i = \sum_{j \in J_i} v_{ij}, \qquad \sum_{i \neq k} \sum_{j \in J_i} t_{ij} z_{ij} \le K. \tag{8}$$

Together with (2) and (1), these form a convex mixed-integer quadratic problem. A declared optional allowance adds its corresponding sum over optional names. Cash has no trade selector.

Every method uses the same prepared state domain. The name Matched Exact identifies this mathematical problem, not a certificate for every time-limited returned vector. The implementation solves an equivalent conic canonical representation. Native status, native incumbents and externally retained weights remain separately attributed.

Fixing one state per instrument yields a convex quadratic program. Fixed coordinates and cash can be eliminated algebraically. Heuristic supports use the same joint-sizing formulation. A conditional fixed-state refinement of an integer incumbent preserves its declared state choices and is audited separately; native outputs are never rounded into acceptance.

5. Ranked Execution

5.1. Cash-funded marginal gains

A change in noncash instrument i is funded through $u_i = e_i - e_k$. At working point w, define

$$g_i = u_i^\top \Sigma (w - m), \qquad s_i = u_i^\top \Sigma u_i \ge 0. \tag{9}$$

A displacement x changes risk by

$$f(w + x u_i) - f(w) = 2 g_i x + s_i x^2. \tag{10}$$

For state displacement interval $D_i = [d_i^-, d_i^+]$, minimise this scalar quadratic. When $s_i > 0$, the minimiser is the projection of $-g_i/s_i$ onto D_i; when $s_i = 0$, an endpoint minimises the linear expression. Ties are resolved deterministically.

The priority is the negative objective change at that minimiser. Greedy selection admits positive-gain optional names within the allowance and updates covariance products after each virtual move:

$$\Sigma(w-m) \leftarrow \Sigma(w-m) + x_i(\Sigma_{:i} - \Sigma_{:k}). \tag{11}$$

A selected name cannot consume another ticket by choosing a second state. Instrument identifiers and state indices break score ties.

Virtual working points are support proposals. They need not satisfy coupled cash or group bounds. Those constraints enter joint sizing, and only an audited jointly sized vector can replace the incumbent. A marginal priority guides computation without becoming an acceptance rule.

5.2. Joint sizing and mandatory coordinates

The search starts from the shared audited incumbent q. A mandatory-only QP first explores movable mandatory coordinates and their bounded holding alternatives. A feasible base can still be improved along these coordinates; an empty optional support must not suppress their optimization.

For a proposed optional support, selected states are free within their intervals; unselected optionals remain at base. Movable mandatory coordinates and cash are jointly sized with them. Funding, instrument bounds and original group rows are included. Every candidate is audited before retention.

The default explores at most 32 mandatory state combinations per support. Truncation is reported. Failure of one assignment does not prove that every assignment for those instrument names is infeasible. A feasible candidate with more risk than the incumbent leaves the incumbent unchanged.

If initial greedy sizing supplies no audited candidate, a bounded rescue list tries constraint-directed admissions, removals and replacements. Its separate allowance is at most 24 support trials. Rescue respects the same cap and states; it cannot introduce a new exemption or broaden a corridor.

5.3. Coordinated state changes

Isolated priorities can miss a useful combination. Funding and group restrictions may make a buy and sale jointly attractive even when their isolated proposals are unsuitable. The method supplements one drop and one exchange pass with a bounded coordinated neighbourhood.

At incumbent $\bar{w}$, let $C_{ij} = u_i^\top \Sigma u_j$. For proposed state-displacement intervals D_i, D_j, the two-coordinate priority is

$$-\min_{x \in D_i,\, y \in D_j} \{2g_i x + 2g_j y + C_{ii}x^2 + 2C_{ij}xy + C_{jj}y^2\}. \tag{12}$$

The positive-semidefinite box quadratic is evaluated through boundary minima and any admissible interior stationary point. This priority is inexpensive relative to complete joint sizing.

The neighbourhood changes one or two optional states: admissions, removals, replacements within available slots, or holding alternatives. Priorities do not include all coupled rows or reoptimisation of the other selected coordinates. Consequently, even a negative isolated priority need not exclude a support from the shortlist.

The best unvisited candidates form a bounded shortlist. Joint sizing evaluates them under all constraints. If the incumbent improves and trials remain, the search restarts from the new incumbent. At most three anchors share a total coordinated budget of 24 support trials. A pass without improvement stops the stage. These are support trials; mandatory alternatives and numerical retries can require more than 24 native QP calls.

A two-state neighbourhood cannot simultaneously replace two selected names with two different new names: that requires four changes. A truncated shortlist can also omit the best feasible pair. Restarts expose additional moves after earlier improvements but establish neither local nor global optimality.

Table 1: Ranked Execution with audited incumbent retention.

Step	Operation
1	Validate inputs, compile states and audit the shared prepared base.
2	Size movable mandatory states without optional trades.
3	Construct cash-funded greedy priorities and jointly size selected states.
4	If needed, attempt the bounded feasibility-rescue shortlist.
5	Attempt one drop and one exchange pass over the initial accepted support.
6	Jointly size coordinated changes; restart after improvement while the shared trial and anchor allowances permit.
7	Return the best audited incumbent with timing and failure diagnostics.

5.4. Properties and computational cost

Proposition 1 (Conditional incumbent retention). *For fixed arrays and acceptance rules, if the initial incumbent passes the return audit and replacement requires an audited strict objective improvement, the returned objective is no worse than that of the initial incumbent.*

PROOF. The initial incumbent is stored. Each replacement passes the same audit and has smaller objective. Failure to find a replacement leaves it unchanged.

This delivery property is equally available to the mixed-integer reference. It is not an approximation ratio or an optimality certificate and says nothing about future risk after rolling holdings diverge.

For dense covariance, the initial covariance-vector product costs $O(N^2)$. With bounded states per instrument, each greedy score and column update costs $O(N)$, giving $O(N^2 + BN)$ arithmetic for that stage and $O(N^2)$ input storage. These counts exclude validation, convex sizing, mandatory enumeration and improvement search.

With s available state changes, coordinated enumeration can inspect $O(s^2)$ pairs per anchor. A bounded shortlist does not make its generation constant work in universe size. Support-key construction and ordering add overhead. At most three anchors share the same sizing budget, while mandatory enumeration multiplies QP counts. A support cap is therefore not a wall-clock guarantee.

The library accepts exact factor-plus-diagonal covariance representations when available. The empirical comparison retains its supplied dense matrices, including residual correlations. Substituting a diagonal residual approximation would change the experiment and is not done here.

6. Numerical acceptance and comparison protocol

6.1. Original-unit audit

Every final vector is checked in NAV fractions against unchanged supplied arrays. Feasibility tolerance is 10^{-7} and actual-trade zero is 10^{-8}. Holding floors and minimum tickets retain their declared values. The independent benchmark audit checks funding, instrument intervals, group exposures, floors, nonexempt ticket sizes and both ticket allowances. It calls neither the implementation's state compiler nor its audit evaluator. Returned vectors are not renormalised or rounded.

Numerical QPs express movable weights in basis points, eliminating fixed coordinates and cash algebraically. In fixed-state sizing, a bound or group row is removed only when every remaining variable coefficient is exactly zero and its constant value satisfies both unchanged endpoints. Nonzero coefficients, however small, and constant contradictions are retained. This

algebraic reduction avoids degenerate constant rows without enlarging the feasible set. Native integer hulls and their relaxations retain their original graph. Group rows use the same bounded residual envelope as the original-unit audit. Original limits remain authoritative at return. CLARABEL uses absolute/relative gap and feasibility controls of 10^{-10}. A numerical solver exception permits one fresh attempt with equilibration disabled, retaining the graph, objective, rows and tolerances. An infeasible status does not trigger that retry.

This aligns numerical construction and acceptance. It does not prove literal feasibility beyond the declared tolerance, or convert a native lower bound into a certified optimality gap. The benchmark records raw statuses and candidate rejection alongside delivered portfolios.

6.2. Five comparison arms

Table 2 specifies the experiment. Common preparation is performed once and fixed across arms. ExactBase starts from the prepared incumbent. ExactRanked is an independent call supplied with Ranked's incumbent. Neither call receives a pooled best result across algorithms.

Table 2: Arms under the same prepared CORRIDOR domain.

Arm	Definition
Ranked	Full default search, with up to three coordinated anchors sharing 24 support trials.
Greedy	Greedy and mandatory/selected-state sizing; no rescue, drop, exchange or coordinated search. The same incumbent is retained.
OneAnchor	Default search with at most one coordinated anchor.
ExactBase	Matched mixed-integer solve started from the prepared base.
ExactRanked	Separate matched solve started from Ranked's audited incumbent.

Both Exact calls use a 10-second wrapper allowance, one native thread, seed zero and relative MIP-gap setting 10^{-4}. Graph construction, start lifting and conditional fixed-state refinement consume this allowance. Native operations can overrun it; measured wall times are reported.

Native audited incumbents, retained external starts and refined candidates have distinct attribution. An ExactRanked delivery equal to Ranked

does not establish independent native discovery. An ExactBase delivery worse than Ranked under a time limit does not mean its mathematical optimum is worse.

Arms run serially on one host. Algorithm-call times exclude imports and input loading. Preparation is reported separately and included in end-to-end time. ExactRanked additionally includes its Ranked start-generation cost. These are declared computational controls, not equal elapsed budgets. The recorded host runs Windows 11 on an Intel Core Ultra 7 265 processor. The cohort uses Python 3.12.14, OP 9.1.0.dev1, QIS 5.44.3, CVXPY 1.9.2, MOSEK 11.2.3 and CLARABEL 0.11.1; the replication protocol records the full numerical package versions and execution-source hashes.

7. The 302-ETF benchmark

7.1. Frozen supplied inputs

The source universe contains 302 publicly traded ETFs, retrospectively selected in October 2026. Frozen model allocations span June 2008 through September 2026. We construct 219 monthly decisions from July 2008 onward. Eligible covariance support expands with available observations. The supplied universe is not a point-in-time investable-universe reconstruction.

Each case begins from the previous supplied model portfolio, normalised once to unit investment and drifted through the intervening month with the canonical portfolio helper. This removes the previous model's small investment residual when defining the hypothetical current book. The current raw target remains unchanged in the risk objective. No competing execution strategy's historical holdings define these cases.

The experiment is conditional on supplied targets and risk estimates. It does not validate upstream model selection, estimate the economic value of signals or establish out-of-sample investment performance. Monthly cases share assets and market history and are not treated as independent samples.

Every instrument has a 10% upper bound, including settlement cash; unavailable instruments have zero upper bound. Exposure bands are 0–10% liquidity, 25–45% fixed income, 35–65% equity and 5–15% alternatives. A mixed-allocation instrument has fractional equity/fixed-income loadings. Monthly eligibility and lifecycle declarations precede common preparation.

Ordinary actual tickets are at least 25 basis points of NAV. Equity and alternative holdings have a 50-basis-point floor; fixed income and liquidity have

no positive floor. The optional allowance is ten tickets plus the frozen mandatory reservation. Compulsory interventions retain their minimum-ticket exemption and remain counted. There is no separate position-count cap.

7.2. Public replication boundary

The inputs contain current and target weights, the pre-preparation base, annual covariance, corridor and full bounds, group rows, eligibility and fixed-coordinate flags, mandatory categories, exemptions and ticket rules. Stable identifiers such as `Asset001` replace instrument labels; `Cash` identifies funding.

There are no raw vendor price histories, private portfolio observations, factor-estimation inputs, local filesystem paths or private identifier mapping in the replication dataset. The supplied arrays reproduce common preparation and all execution arms. Model targets alone would not suffice.

Opaque identifiers provide de-identification, not a guarantee that numerical histories cannot be recognised. The benchmark concerns publicly traded instruments and makes no claim to conceal their market characteristics. Stable labels and unchanged coordinate order avoid conflating renaming with algorithm changes.

Every data file is hashed before loading. Arrays contain no Python pickle objects. Reference outputs are used only for verification and reporting, never to select a support or improve an incumbent. Replication calls reusable OP execution and QIS risk analytics without a private application or an upstream research repository.

7.3. Results and interpretation

All 219 decisions were attempted. Common preparation succeeded on 219; all five arms supplied independently audited portfolios on 219 decisions. The archive retains every attempted case, including any failed preparation. Comparisons use paired audited outputs.

Table 3 uses the common audited subset. Tracking error is annual basis points; computation is elapsed seconds. The prepared base has mean tracking error 32.892 basis points. Median common preparation takes 0.086 seconds.

Pairwise differences use a descriptive equality band of 10^{-4} basis points, distinct from the portfolio feasibility tolerance. Negative differences favour Ranked. No independence-based significance test is applied to these serially related decisions.

Table 3: Conditional execution quality and measured computation. End-to-end (E2E) includes common preparation and, for ExactRanked, Ranked start generation.

Arm	Mean TE	Median TE	Median s	95th pct. s	Median E2E s
Ranked	16.825	4.607	0.095	1.203	0.416
Greedy	17.077	4.712	0.081	0.736	0.303
OneAnchor	16.825	4.607	0.092	0.894	0.414
ExactBase	16.825	4.607	0.064	0.145	0.164
ExactRanked	16.825	4.607	0.061	0.189	0.479

Table 4: Ranked relative to each comparator on paired audited decisions. The last column is mean Ranked minus comparator TE, in basis points.

Comparator	Pairs	Ranked lower	Equal	Ranked higher	Mean difference
Greedy	219	9	210	0	-0.25135
OneAnchor	219	0	219	0	0.00000
ExactBase	219	9	208	2	-0.00004
ExactRanked	219	0	219	0	0.00000

Optional candidate counts range from 0 to 5 (median 1); 107 prepared cases have no optional candidate. Mandatory reservations range from 8 to 71 (median 30). These are the effective search dimensions, not the nominal 302-asset universe.

Preparation retains 65 already audited bases, repairs 79 within the ordinary corridor, and uses FULL preparation on 75 decisions. The resulting pins and corridor exceptions are shared by every arm.

The optional ten-ticket allowance never binds in this sample: every prepared case has at most five optional candidates. Mandatory interventions dominate the ticket count. Thus these data do not test scarce-ticket selection among a large optional candidate set.

Ranked improves on Greedy in 9 cases and is equal in 210. OneAnchor and the full search have equal delivered objectives on 219 paired cases. ExactBase median call time is 0.064 seconds versus Ranked's 0.095. The measured computation does not support a general heuristic speed advantage.

ExactBase delivered-source attribution is *boundary refinement*: 1, *common prepared incumbent*: 72, *native callback*: 132, *native return*: 14. It recorded a native audited candidate on 219 of 219 prepared cases. A callback or native return may reproduce a submitted start; this count alone is

not evidence of an independently discovered improvement.

ExactRanked delivered-source attribution is *common prepared incumbent*: 72, *native callback*: 55, *native return*: 7, *requested mip start*: 85. It recorded a native audited candidate on 219 of 219 prepared cases. A callback or native return may reproduce a submitted start; this count alone is not evidence of an independently discovered improvement.

The largest delivered TE disadvantage of Ranked against ExactRanked is 0.00004 basis points (case 0046). Delivered differences compare feasible incumbents, not certified global gaps. Full weights, numerical audits, native statuses and timing are retained per decision.

This frozen benchmark tests a distinct public-input cohort, but its low effective candidate counts limit conclusions about dense optional search. The comparison establishes the reported conditional outcomes; it does not establish universal superiority or realised investment gains.

An independent sizing check uses case 0027, with 17 mandatory reservations and no optional candidates. The admissible mandatory state has one movable noncash coordinate. Intersecting its instrument, cash and group intervals and projecting the scalar quadratic's stationary point gives tracking error 79.64086 basis points. Ranked agrees with that independently reconstructed solution to 2.4×10^{-14} in maximum absolute weight difference. The alternative fixed state remains infeasible. The check verifies conditional sizing, not global performance on other support patterns.

8. Limitations and conclusion

Ranked Execution separates cash-funded prioritisation from the joint sizing needed for coupled constraints. Executable states encode minimum trades and holding alternatives; mandatory declarations make counting and exemptions explicit. Coordinated search broadens the combinations considered under a fixed domain, and original-unit audits control delivery.

A better objective than a time-limited mixed-integer call is not a global certificate. An Exact improvement identifies a search or numerical limit that must remain visible. Retained starts, native discoveries and numerical refinements are different sources of delivered portfolios.

The empirical limits include the retrospective universe, supplied rather than refitted estimates, dependent monthly cases and the number of genuinely optional states after preparation. A large nominal universe does not

by itself establish difficult sparse search. The retained candidate counts measure the scale actually tested. Numerical corrections were informed by these instances; the sample is not an untouched validation cohort.

This benchmark does not simulate every algorithm's recursively evolving holdings. Such a study requires asset returns and a separate implementation convention; its outcomes combine local decisions with path dependence. No realised return, Sharpe ratio or trading-cost advantage follows from the conditional comparison.

The portable instance contract makes these limits testable. Subsequent studies can vary ticket budgets, floors, group tightness and candidate density under frozen protocols, and compare established methods on the same trade domain. The present contribution is a concrete, auditable execution heuristic with explicit controls and a reproducible matched benchmark.

Data and code availability

The accompanying package contains de-identified numerical decision inputs, checksums, the five-arm runner, independent audits and table-generation code. The reusable solver is in OptimalPortfolios. Tracking-error reporting uses QIS. This public-facing draft is prepared for release; a persistent dataset identifier and public release URL have not yet been assigned. Execution conditional on supplied estimates is reproduced; upstream allocation-model estimation and raw market-data histories are outside the release.

Author declarations

Affiliation, competing-interest and final authorship statements require author completion before submission. AI assistance was used in implementation, analysis and drafting; the author must review and approve the final disclosure.

References

Bertsimas, D., Cory-Wright, R., 2022. A scalable algorithm for sparse portfolio selection. INFORMS Journal on Computing 34(3), 1489–1511. doi:10.1287/ijoc.2021.1127.

Guastaroba, G., Speranza, M.G., 2012. Kernel Search: An application to the index tracking problem. European Journal of Operational Research 217(1), 54–68. doi:10.1016/j.ejor.2011.09.004.

Mutunge, P., Haugland, D., 2018. Minimizing the tracking error of cardinality constrained portfolios. Computers & Operations Research 90, 33–41. doi:10.1016/j.cor.2017.09.002.

Strub, O., Baumann, P., 2018. Optimal construction and rebalancing of index-tracking portfolios. European Journal of Operational Research 264(1), 370–387. doi:10.1016/j.ejor.2017.06.055.